

claude-windows / 7d6bbb83-42c0-459b-9e23-e14e96d4cb2d

Metadata

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- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\7d6bbb83-42c0-459b-9e23-e14e96d4cb2d\subagents\workflows\wf_1dc2acd6-052\agent-a328cf2f8499d01d0.jsonl`  
- SHA-256: `e2097a22083862df500d850acc8de207251afef1ca16349f9a8c3eeef89bfcc9`  
- Source modified: `2026-06-13T20:24:35+00:00`  
- Imported at: `2026-07-08T16:00:13+00:00`  
- project: `wf_1dc2acd6-052`  
- session_id: `7d6bbb83-42c0-459b-9e23-e14e96d4cb2d`
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Transcript

1. user (2026-06-13T20:19:31.347Z)

You are an ADVERSARIAL fact-checker. Theme: "Multi-scale / hierarchical temporal modeling; video-level context & phase/stage segmentation".

Here are a teammate's survey findings (JSON):

```
{"theme": "Multi-scale / hierarchical temporal modeling; video-level context & phase/stage segmentation", "our_pattern": "A three-tier temporal SCOPE taxonomy for rally detection. (1) frame-scope DETECTORS produce expensive per-frame signals (shuttle x,y,visibility from a frozen WASB/TrackNet CNN; person boxes/tracks; audio thwack; a proposed per-frame shuttle-state classifier), persisted once. (2) window-scope ANALYZERS emit cheap scale/translation-invariant aggregate features per candidate rally window (net-crossing count, service-fault halt-in-net, person two-sidedness). (3) session-scope ANALYZERS run over the WHOLE timeline to emit labeled spans + per-window membership features ? notably WARM-UP/practice detection (a long continuous-rally stretch lacking the point/dead-time cadence) that CONDITIONS the per-window rally decision. Every cue is a (value, confidence) feature column consumed by a small logistic-regression scorer; coarse session-level state gates/contextualizes fine-grained per-window decisions. Evaluated by temporal IoU vs scarce vendor golden intervals, with leave-one-video-out (group-by-match) splits.", "key_papers": [{"title": "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation", "authors": "Yazan Abu Farha, Juergen Gall", "year": "2019", "venue": "CVPR 2019", "url_or_id": "arXiv:1903.01945", "what_it_does": "Multi-stage architecture for temporal action segmentation: each stage uses dilated temporal convolutions over the full-resolution timeline to make an initial frame-wise prediction that the next stage REFINES. Adds a smoothing loss (truncated MSE over log-prob differences) explicitly to penalize over-segmentation. Reports F1@{10,25,50}, segmental edit score, frame accuracy on 50Salads/GTEA/Breakfast.", "overlap": "partial", "overlap_explanation": "Same core problem (segment an untrimmed timeline into action vs background spans, fight over-segmentation) and same canonical metrics we should adopt (F1@k overlap, segmental edit score) ? these are richer than raw temporal IoU. The iterative refinement IS conceptually our reconciler pass (initial decision -> refined decision using a wider temporal view). Differs in that MS-TCN is one end-to-end deep net learning everything from frame features, whereas we deliberately use hand-crafted multi-scope feature columns + a tiny logistic regression because labels are scarce; and our refinement is a report-first, A/B-gated rule pass, not a learned stage."}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF)", "authors": "Yuchi Ishikawa, Seito Kasai, Yoshimitsu Aoki, Hirokatsu Kataoka", "year": "2021", "venue": "WACV 2021", "url_or_id": "arXiv:2007.06866", "what_it_does": "Splits action segmentation into two branches over a shared long-term feature extractor: an Action Segmentation Branch (per-frame class) and a separate class-agnostic Boundary Regression Branch. The predicted boundaries refine/correct the segmentation as a post-process, sharply cutting over-segmentation.", "overlap": "strong", "overlap_explanation": "This is the closest analogue to our RECONCILERS. ASRF treats boundary correction as a SEPARATE, explicit post-hoc pass that reconciles an initial segmentation against an independent boundary signal ? exactly our 'detect where the decision contradicts the evidence, then truncate/split/tighten boundaries.' Our truncate-on-shuttle-inactive and split-on-internal-dead-gap are rule-based instances of ASRF's learned boundary-refinement idea. Differs: ASRF learns boundaries with a neural branch and applies them unconditionally; ours are evidence-driven rules that are report-first and A/B-gated
```

before auto-apply."}, {"title": "Generic play-break event detection for summarization and hierarchical sports video analysis", "authors": "Dian Tjondronegoro, Yi-Ping Phoebe Chen, Binh Pham", "year": "2003", "venue": "IEEE ICME 2003", "url_or_id": "IEEE Xplore document 1220881", "what_it_does": "A single generic algorithm segments a sports broadcast (football, tennis, basketball, soccer) into 'play' vs 'break' intervals using shot-based cinematic features (shot type, shot length), then uses those play/break spans as a HIERARCHICAL outer layer that defines the segments-of-interest fed to finer event detectors (e.g. goal detection keys off the break that follows a goal play).", "overlap": "strong", "overlap_explanation": "Directly validates our session/window scope hierarchy: a coarse, whole-timeline play-vs-break (? rally vs dead-time) layer determines where and how fine-grained detection runs. Our warm-up/practice session-scope span that conditions per-window rally decisions is the same coarse-state-gates-fine-detection move, two decades old in sports video. Differs: they rely on broadcast/cinematic editing cues (replays, shot cuts) that DON'T exist in our static tripod footage; our coarse layer must come from gameplay-cadence signals instead. The play/break terminology and the hierarchical-context framing are ours to adopt."}, {"title": "Global-local multi-stage temporal convolutional network for cataract surgery phase recognition (GL-MSTCN)", "authors": "Lixin Fang et al.", "year": "2022", "venue": "BioMedical Engineering Online (PMC9710114)", "url_or_id": "doi:10.1186/s12938-022-01048-w", "what_it_does": "A multi-stage TCN with explicit GLOBAL and LOCAL streams: local blocks capture fine-grained per-frame/short-window surgical action; a global stream consolidates long-range, video-level temporal context so that locally ambiguous, visually-similar phases are disambiguated by where they sit in the procedure. Reports ~95.8%/96.5% accuracy on cataract datasets.", "overlap": "strong", "overlap_explanation": "Squarely our theme: an explicit global (video-level) context branch improving local (frame/window) decisions for monotonic-phase video. Surgical phase recognition is the canonical literature for 'coarse video-level state conditions fine-grained recognition' and is methodologically transferable to our warm-up-vs-match conditioning. Differs: surgical phases follow a near-fixed order (strong sequence priors via dilated TCN / state-space models), whereas our rallies are repetitive and order-free, so we get less from sequence priors and more from cadence/membership features. Confirms terminology ('global-local', 'phase recognition') and the idea of a learned global-context feature."}, {"title": "Hierarchical Multi-Stage Transformer Architecture for Context-Aware Temporal Action Localization (PCL-Former)", "authors": "(authors per arXiv listing ? not individually verified)", "year": "2025", "venue": "arXiv preprint", "url_or_id": "arXiv:2507.06411", "what_it_does": "A hierarchical, multi-stage transformer that decomposes temporal action localization into a Proposal stage (candidate segments), a Classification stage, and a Localization stage that refines boundaries ? explicitly separating proposal generation, scoring, and boundary refinement with multi-level (snippet + proposal + video) context.", "overlap": "partial", "overlap_explanation": "Mirrors our DECOUPLED pipeline: candidate windowing -> scoring -> boundary refinement, with multi-granularity context ? the same separation-of-concerns we encode as windowing + logistic scorer + reconcilers. Marked partial/weaker because I verified it only from the search snippet and arXiv listing (not the full PDF), it is a heavy deep transformer rather than our lightweight scorer, and authorship was not independently confirmed."}, {"title": "LiteEval: A Coarse-to-Fine Framework for Resource Efficient Video Recognition", "authors": "Zuxuan Wu et al.", "year": "2019", "venue": "NeurIPS 2019", "url_or_id": "arXiv:1912.01601", "what_it_does": "A coarse LSTM glimpses cheap low-res features over a video, and a conditional GATING module decides whether each frame warrants expensive fine-grained feature computation ? i.e. a coarse global pass gates where expensive fine processing runs.", "overlap": "partial", "overlap_explanation": "Validates the efficiency half of our DETECTION?DECISION + scope split: run cheap coarse analysis first, spend expensive fine compute only where the coarse layer says it matters. Differs: LiteEval's gating is a learned, online, per-frame compute-saving decision; our split is offline persist-once/re-score-forever where the expensive detector ALWAYS runs once and the cheap re-scoring is what varies. So the framing transfers but our motivation (re-scorability + scarce labels) differs from theirs (inference FLOPs)."}], "established_terms": ["temporal action segmentation (frame-wise labeling of an untrimmed timeline into action/background spans ? the formal name for our rally/dead-time split)", "play-break detection (the sports-video term for rally-vs-dead-time; our session/window coarse layer)", "surgical/cooking PHASE recognition (the canonical literature for coarse video-level state conditioning fine recognition)", "global-local / multi-scale temporal modeling (explicit global video-level branch + local frame branch)", "over-segmentation error (the named failure our reconcilers and a smoothing loss fight)", "action boundary regression / boundary refinement (ASRF ? the formal name for our boundary-tightening/truncate/split reconcilers)", "temporal proposal generation -> classification -> localization (the decoupled three-stage pipeline; our windowing -> scorer -> reconcile)", "leave-one-subject-out / leave-one-video-out cross-validation (our LOVO; split-by-group to prevent leakage)", "F1@k (overlap F1 at IoU thresholds 10/25/50) and segmental edit score (the standard segmentation metrics beyond raw

temporal IoU)], "techniques_to_adopt": [{"technique": "Report F1@{10,25,50} and segmental edit score alongside temporal IoU.", "why": "These are the field-standard metrics for temporal action segmentation (MS-TCN, ASRF). Raw temporal IoU rewards frame-overlap but is nearly blind to over-segmentation (a span chopped into 3 pieces can still score high IoU); F1@k and edit score directly penalize fragmentation and boundary errors ? exactly the failures our reconcilers target. Adopting them makes our 'lift' measurements comparable to the literature and catches regressions IoU hides.", "from_paper": "MS-TCN (Abu Farha & Gall, 2019); ASRF (Ishikawa et al., 2021)", "effort": "low"}, {"technique": "Frame the reconcilers explicitly as a class-agnostic boundary-refinement stage (ASRF pattern): an independent boundary signal corrects an initial segmentation as a post-process.", "why": "ASRF shows that separating 'is this a rally' from 'where exactly are its boundaries' and refining boundaries against an independent signal yields large drops in over-segmentation. Our truncate-on-inactive / split-on-internal-gap / tighten-to-serve-contact rules ARE this stage; adopting the framing (and ASRF's boundary metric) gives us a principled, citable home and a way to measure boundary quality separately from detection quality.", "from_paper": "ASRF (Ishikawa et al., 2021)", "effort": "medium"}, {"technique": "Add a smoothing/anti-fragmentation regularizer or post-filter to the decision, not just rely on reconcilers.", "why": "MS-TCN's truncated-MSE smoothing loss is the canonical fix for over-segmentation. Since our scorer is a logistic regression over windows (not a frame net), the analogue is a cheap merge/min-duration/transition-penalty post-filter or a HMM/Viterbi smoothing over the per-window rally probabilities ? a learned, measured smoother rather than only rule-based reconcilers, reducing reliance on hand-tuned rule precedence.", "from_paper": "MS-TCN (Abu Farha & Gall, 2019)", "effort": "medium"}, {"technique": "Treat the session-scope warm-up/play-break layer as an explicit hierarchical outer segmentation whose spans key the inner detector (Tjondronegoro play-break scheme).", "why": "The play-break work establishes the exact hierarchy we are building (coarse play/break spans define segments-of-interest for finer detectors) as a recognized design, giving us prior-art cover and the 'play/break' vocabulary. It also flags the right inputs: since we lack broadcast cinematic cues, our coarse layer must be built from gameplay-cadence features (rally/dead-time rhythm), which is a concrete design constraint to encode.", "from_paper": "Generic play-break event detection (Tjondronegoro et al., 2003)", "effort": "medium"}, {"technique": "Model the global/session context as a learned membership FEATURE per window (global-local fusion), and verify it actually helps via ablation.", "why": "GL-MSTCN shows a global stream disambiguates locally-similar segments ? our intended warm-up-conditions-rally move. Encoding session-scope state as an extra feature column (not a hard pre-filter) keeps it inside the 'signal not verdict' regime and lets the scorer learn its weight; the surgical literature's ablations (global vs local-only) are the template for proving the global cue earns its place.", "from_paper": "GL-MSTCN (Fang et al., 2022)", "effort": "low"}], "pitfalls": ["Over-segmentation is THE dominant failure mode of timeline segmentation (MS-TCN, ASRF both built specifically to fight it). Our merge/split reconcilers must be measured on F1@k/edit, not IoU, or fragmentation regressions will slip through invisibly.", "Coarse-state gating can hard-cut recall: if the session-scope warm-up/play-break layer is wrong, it suppresses real rallies inside a mislabeled span (the classic hierarchical-gating risk where an upstream error is unrecoverable downstream). Keep session context as a soft feature column, never a hard pre-filter, so a wrong coarse call is recoverable by the scorer ? consistent with your 'signal not verdict' rule.", "LOVO leakage is subtle: the action-segmentation field splits by SUBJECT/participant, not just by clip, because frames from the same person/session share idiosyncrasies. Your group-by-match LOVO is correct, but watch second-order leakage ? features fit/normalized or thresholds tuned on the full corpus, or a shared vendor-labeling bias, still leak across folds.", "Static-tripod transfer gap: the strongest sports-video priors (play-break, badminton broadcast pipelines) lean on broadcast editing cues ? replays, shot cuts, score graphics ? that DON'T exist in your footage. Borrowing their methods wholesale will silently fail; you must re-source the coarse cue from gameplay cadence, and the literature won't have done that for you.", "Tiny-corpus overfitting of the gating/global cue: with ~6 matches, a session-scope feature can latch onto match-specific quirks (one warm-up style) and look like lift in-fold while failing out-of-fold. Surgical/segmentation papers train on dozens-to-hundreds of videos; insist on per-fold LOVO lift for any global feature before promotion, exactly as your harness specifies.", "Sequence-prior mismatch: MS-TCN/surgical models gain much of their power from near-fixed phase ORDER (Viterbi/grammar priors). Badminton rallies are repetitive and order-free, so importing strong sequence-transition priors can hurt; prefer cadence/duration/membership features over learned action-grammar transitions."], "verdict": "partial-overlap", "verdict_rationale": "The structural pattern ? a coarse, video/session-level state (play-break, surgical phase, warm-up) conditioning fine-grained per-window decisions, with a decoupled propose -> score -> boundary-refine pipeline ? is well-established across three mature literatures (temporal action segmentation: MS-TCN/ASRF; sports play-break: Tjondronegoro 2003; global-local phase recognition: GL-MSTCN). So the hierarchical-context idea itself is NOT novel and we should cite and borrow vocabulary and

metrics from it rather than reinvent. What is genuinely differentiated (and not reinventing) is the engineering regime: persist-once/re-score-forever detection?decision, every cue as a calibrated (value,confidence) feature column with a deliberately tiny logistic-regression scorer for a ~6-match scarce-label corpus, report-first A/B-gated reconcilers instead of unconditional learned refinement, and a no-regression flag-gated harness ? a combination the cited deep, data-hungry, end-to-end methods do not provide. Hence partial-overlap: adopt their terminology (play-break, phase recognition, over-segmentation, boundary refinement), their metrics (F1@k, edit score), and their global-local ablation discipline as course corrections, while keeping our lightweight, re-scorable, regression-safe architecture as the distinctive contribution. Two caveats lower confidence on the edges: GL-MSTCN author detail and PCL-Former authorship were not fully verified from primary PDFs (marked accordingly), and the closest 'pure' analogue to our exact scope-taxonomy-as-feature-columns design (vs end-to-end nets) was not found verbatim, leaving a thin slice that may be a small genuine gap.", "search_queries_used": ["hierarchical multi-granularity temporal action localization video-level context", "surgical phase recognition multi-stage temporal model global context", "MS-TCN multi-stage temporal convolutional network action segmentation over-segmentation", "global video context improves frame-level action recognition coarse-to-fine gating", "tennis badminton broadcast video play break rally segmentation temporal", "sports video event detection hierarchical game phase structure prior localization", "video temporal segmentation warm-up practice versus match detection sports scene classification", "Tjondronegoro generic play-break event detection summarization hierarchical sports video", "leave-one-video-out cross-validation temporal action segmentation small dataset leakage overfitting sports", "Alleviating over-segmentation action segmentation boundary refinement ASRF separate boundary regression", "two-stage cascade video-level scene gating frame-level detection conditional structured prediction temporal IoU"]}]

Your job is to be skeptical and protect against two failure modes: (1) FABRICATED citations, (2) OVERCLAIMED novelty.

DO, using WebSearch/WebFetch (load via ToolSearch "select:WebSearch,WebFetch" if needed):

- For each cited paper, search its EXACT title (+authors) to confirm it is a REAL paper. Mark `verified_real` yes/no/uncertain and flag any that don't resolve in `fabrication_flags`. Be strict: a plausible-sounding title that you cannot find is a fabrication flag.
- For every "novel-combination" or "genuine-gap" verdict, actively try to REFUTE it: run fresh searches for the CLOSEST prior art that would downgrade the verdict toward "reinventing"/"partial-overlap". Default to assuming prior art exists until your searches come up empty. List any `closer_prior_art` you find (with real ids).
- Give a `corrected_verdict` + rationale. Add `extra_course_corrections` if your searches surfaced adopt/avoid notes.

Return the structured object.

2. assistant (2026-06-13T20:19:33.870Z)

I'll fact-check this survey adversarially. Let me load the web tools and verify each citation, then hunt for closer prior art.

3. assistant (2026-06-13T20:19:38.066Z)

Starting verification. Let me search for the citations in parallel, beginning with the ones most likely to be problematic (the 2003 IEEE paper, the 2025 arXiv preprint with unverified authors, and GL-MSTCN).

4. user (2026-06-13T20:19:44.889Z)

Web search results for query: "Alleviating Over-segmentation Errors Detecting Action Boundaries ASRF Ishikawa Kasai Aoki Kataoka"

Links: [{"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://yiskw713.github.io/asrf/"}, {"title": "[PDF] Alleviating Over-segmentation Errors by Detecting Action Boundaries | Semantic"}]

Scholar", "url": "https://www.semanticscholar.org/paper/Alleviating-Over-segmentation-Errors-by-Detecting-Ishikawa-Kasai/4fed0235614df46fa0c53e05b71e9939b53a1bdf"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://openaccess.thecvf.com/content/WACV2021/papers/Ishikawa_Alleviating-Over-Segmentation_Errors_by_Detecting_Action_Boundaries_WACV_2021_paper.pdf"}, {"title": "GitHub - yiskw713/asrf: Code for 'Alleviating Over-segmentation Errors by Detecting Action Boundaries' accepted in WACV2021", "url": "https://github.com/yiskw713/asrf"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/9423184/"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries | Request PDF", "url": "https://www.researchgate.net/publication/352386005_Alleviating-Over-segmentation_Errors_by_Detecting_Action_Boundaries"}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries", "url": "https://arxiv.org/pdf/2007.06866"}, {"title": "Alleviating Over-Segmentation Errors by Detecting Action Boundaries", "url": "https://papertalk.org/papertalks/27315"}, {"title": "Permutation-Aware Action Segmentation via Unsupervised Frame-to-Segment Alignment", "url": "https://arxiv.org/pdf/2305.19478"}]

Based on the search results, here's what I found:

This is a paper by Yuchi Ishikawa, Seito Kasai, Yoshimitsu Aoki, and Hirokatsu Kataoka titled "Alleviating Over-segmentation Errors by Detecting Action Boundaries," published in WACV 2021.

The paper proposes an Action Segment Refinement Framework (ASRF) consisting of a long-term feature extractor and two branches: the Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB).

The ASB classifies video frames with action classes, while the BRB regresses action boundary probabilities, and the action boundaries predicted by the BRB refine the output from the ASB to achieve significant performance improvement.

The ASB and the BRB are complementary to each other, which enables the reduction of over-segmentation errors.

The official code repository for this work is available on GitHub at <https://github.com/yiskw713/asrf>, and the paper is accessible through IEEE Xplore, arXiv, and the WACV 2021 proceedings.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-06-13T20:19:48.411Z)

Web search results for query: "MS-TCN Multi-Stage Temporal Convolutional Network Action Segmentation Abu Farha Gall CVPR 2019"

Links: [{"title": "GitHub - yabufarha/ms-tcn · GitHub", "url": "https://github.com/yabufarha/ms-tcn"}, {"title": "Abu Farha MS-TCN Multi-Stage Temporal Convolutional Network For Action Segmentation CVPR 2019 Paper | PDF | Image Segmentation | Systems Science", "url": "https://www.scribd.com/document/487365349/Abu-Farha-MS-TCN-Multi-Stage-Temporal-Convolutional-Network-for-Action-Segmentation-CVPR-2019-paper"}, {"title": "[PDF] MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/MS-TCN:-Multi-Stage-Temporal-Convolutional-Network-Farha-Gall/b4673e744d0ded47fe6df3b6314f79a41359578b"}, {"title": "Abu Farha MS-TCN Multi-Stage Temporal Convolutional Network For Action Segmentation CVPR 2019 Paper | PDF", "url": "https://www.scribd.com/document/648823956/Abu-Farha-MS-TCN-Multi-Stage-Temporal-Convolutional-Network-for-Action-Segmentation-CVPR-2019-paper"}, {"title": "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation [Quick Review]", "url": "https://liner.com/review/mstcn-multistage-temporal-convolutional-network-for-action-segmentation"}, {"title": "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/8953830/"}, {"title": "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation -

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ADS", "url": "https://ui.adsabs.harvard.edu/abs/2019arXiv190301945A/abstract"}, {"title": "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation", "url": "https://arxiv.org/pdf/1903.01945"}, {"title": "Frontal Low-rank Random Tensors for Fine-grained Action Segmentation", "url": "https://arxiv.org/pdf/1906.01004"}, {"title": "(PDF) MS-TCN++: Multi-Stage Temporal Convolutional Network for Action Segmentation", "url": "https://www.researchgate.net/publication/344127554_MS-TCN_Multi-Stage_Temporal_Convolutional_Network_for_Action_Segmentation"}]]
```

I've found comprehensive information about the MS-TCN paper. Here are the key findings:

Paper Overview

The paper "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation" was authored by Y. Abu Farha and J. Gall and published at the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) in 2019.

Key Contributions

The paper introduces a multi-stage architecture for the temporal action segmentation task, where each stage features a set of dilated temporal convolutions to generate an initial prediction that is refined by the next one.

The architecture is trained using a combination of a classification loss and a proposed smoothing loss that penalizes over-segmentation errors.

Results

The multi-stage architecture achieves state-of-the-art results on three challenging datasets: 50Salads, Georgia Tech Egocentric Activities (GTEA), and the Breakfast dataset.

Motivation

Temporally locating and classifying action segments in long untrimmed videos is of particular interest to many applications like surveillance and robotics.

The paper is widely cited and has become influential in the field, with variations like MS-TCN++ being developed subsequently.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-13T20:19:51.451Z)

Both MS-TCN (arXiv:1903.01945, CVPR 2019) and ASRF (arXiv:2007.06866, WACV 2021) verify cleanly with correct titles, authors, venues, and IDs. Now the riskier ones.

7. user (2026-06-13T20:19:56.759Z)

Web search results for query: "Tjondronegoro "play-break" event detection summarization hierarchical sports video analysis ICME 2003"

Links: [{"title": "(PDF) Sports video summarization using highlights and play-breaks", "url": "https://www.academia.edu/1172412/Sports_video_summarization_using_highlights_and_play_breaks"}, {"title": "Knowledge-discounted event detection in sports video | IEEE Transactions on Systems, Man, and Cybernetics, Part A: Systems and Humans", "url": "https://dl.acm.org/doi/10.1109/tsmca.2010.2046729"}, {"title": "(PDF) Highlights for more complete sports video summarization", "url": "https://www.researchgate.net/publication/3338768_Highlights_for_more_complete_sports_video_summarization"}, {"title": "Sports Video Summarisation Using Highlights and Play-Breaks | QUT ePrints", "url": "https://eprints.qut.edu.au/1040/"}, {"title": "A scalable and extensible segment-event-object-based sports video retrieval system | ACM Transactions on Multimedia Computing,"}]

Communications, and
Applications", "url": "https://dl.acm.org/doi/10.1145/1352012.1352017"}, {"title": "(PDF)
Integrating Highlights for More Complete Sports Video Summarization", "url": "https://www.academia
.edu/9128449/Integrating_Highlights_for_More_Complete_Sports_Video_Summarization"}, {"title": "(PD
F) Knowledge-discounted event detection in sports video", "url": "https://www.academia.edu/1172422
/Knowledge_discounted_event_detection_in_sports_video?f_ri=49341"}, {"title": "[PDF] Knowledge-
Discounted Event Detection in Sports Video | Semantic
Scholar", "url": "https://www.semanticscholar.org/paper/Knowledge-Discounted-Event-Detection-in-
Sports-Tjondronegoro-Chen/0cca81f3b2928f6d4005e1a8b617960e2fb14d3e"}, {"title": "(PDF) Generic
play-break event detection for summarization and hierarchical sports video
analysis", "url": "https://www.researchgate.net/publication/4027839_Generic_play-
break_event_detection_for_summarization_and_hierarchical_sports_video_analysis"}]

Based on the search results, I found relevant information about this research. Here's what I found:

This paper proposes a single generic real-time (or near real-time) play-break event detection algorithm for multiple sports, which include football, tennis, basketball, and soccer, using shot-based generic cinematic features, such as shot type and shot length.

The detected play-break events are employed for two purposes: 1) all plays in certain sports, such as football and tennis, are presented as summaries, and 2) play-break events, as part of a hierarchical event detection scheme, determine the segments-of-interest for the other event detection algorithms.

An example of such event detection algorithms is given for soccer goal events, where the proposed soccer goal detection algorithm exploits the common cinematic techniques that are employed during the breaks that follow the goal plays.

The paper was published at the Multimedia and Expo conference (ICME '03) in 2003.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-06-13T20:21:15.760Z)

Web search results for query: "Global-local multi-stage temporal convolutional network cataract surgery phase recognition GL-MSTCN BioMedical Engineering OnLine"

Links: [{"title": "Global?local multi-stage temporal convolutional network for cataract surgery phase recognition", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9710114/"}, {"title": "Global?local multi-stage temporal convolutional network for cataract surgery phase recognition | BioMedical Engineering OnLine | Full Text", "url": "https://biomedical-engineering-online.biomedcentral.com/articles/10.1186/s12938-022-01048-w"}, {"title": "(PDF) Global?local multi-stage temporal convolutional network for cataract surgery phase recognition", "url": "https://www.researchgate.net/publication/365890153_Global-local_multi-stage_temporal_convolutional_network_for_cataract_surgery_phase_recognition"}, {"title": "Global-local multi-stage temporal convolutional network for cataract surgery phase recognition - PubMed", "url": "https://pubmed.ncbi.nlm.nih.gov/36451164/"}, {"title": "Phase recognition in manual Small-Incision cataract surgery with MS-TCN + + on the novel SICS-105 dataset | Scientific Reports", "url": "https://www.nature.com/articles/s41598-025-00303-z"}]

API Error: The socket connection was closed unexpectedly. For more information, pass `verbose: true` in the second argument to fetch()

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-13T20:21:23.514Z)

Both verify. Tjondronegoro 2003 "Generic play-break event detection for summarization and hierarchical sports video analysis" (ICME '03) is real with the exact title and described content. GL-MSTCN verifies: doi:10.1186/s12938-022-01048-w, PMC9710114, BioMedical Engineering

OnLine 2022. The description matches (global/local streams for cataract surgery phase recognition).

Now the two flagged-as-uncertain ones: the 2025 PCL-Former arXiv paper (authors not verified) and LiteEval. Let me check the arXiv ID directly and search LiteEval.

10. user (2026-06-13T20:21:29.595Z)

Web search results for query: "LiteEval Coarse-to-Fine Framework Resource Efficient Video Recognition Zuxuan Wu NeurIPS 2019"

Links: [{"title": "LiteEval: A Coarse-to-Fine Framework for Resource Efficient Video Recognition", "url": "https://proceedings.neurips.cc/paper/2019/hash/bd853b475d59821e100d3d24303d7747-Abstract.html"}, {"title": "[1912.01601v1] LiteEval: A Coarse-to-Fine Framework for Resource Efficient Video Recognition", "url": "https://arxiv.org/abs/1912.01601v1"}, {"title": "[1912.01601] LiteEval: A Coarse-to-Fine Framework for Resource Efficient Video Recognition", "url": "https://arxiv.org/abs/1912.01601"}, {"title": "LiteEval: A Coarse-to-Fine Framework for Resource Efficient Video Recognition | DeepAI", "url": "https://deepai.org/publication/liteeval-a-coarse-to-fine-framework-for-resource-efficient-video-recognition"}, {"title": "LiteEval: A Coarse-to-Fine Framework for Resource Efficient Video Recognition", "url": "https://arxiv.org/pdf/1912.01601"}, {"title": "LiteEval: a coarse-to-fine framework for resource efficient ...", "url": "https://dl.acm.org/doi/10.5555/3454287.3454986"}, {"title": "A Coarse-to-Fine Framework for Resource Efficient Video Recognition | International Journal of Computer Vision | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s11263-021-01508-1"}, {"title": "[PDF] A Coarse-to-Fine Framework for Resource Efficient Video Recognition | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/LiteEval:-A-Coarse-to-Fine-Framework-for-Resource-Wu-Xiong/92020e4a2ed14a5f3fd421bcf1111dd2403a237a"}, {"title": "LiteEval: A Coarse-to-Fine Framework for Resource Efficient Video Recognition", "url": "http://papers.neurips.cc/paper/8993-liteeval-a-coarse-to-fine-framework-for-resource-efficient-video-recognition.pdf"}, {"title": "LiteEval: A Coarse-to-Fine Framework for Resource Efficient Video Recognition | Request PDF", "url": "https://www.researchgate.net/publication/337729575_LiteEval_A_Coarse-to-Fine_Framework_for_Resource_Efficient_Video_Recognition"}]

I found the paper you're looking for! Here are the search results:

LiteEval: A Coarse-to-Fine Framework for Resource Efficient Video Recognition is a paper by Zuxuan Wu, Caiming Xiong, Yu-Gang Jiang, and Larry S. Davis published at NeurIPS 2019.

Overview:

The paper presents LiteEval, a simple yet effective coarse-to-fine framework for resource efficient video recognition, suitable for both online and offline scenarios.

Key Technical Approach:

The framework exploits decent yet computationally efficient features derived at a coarse scale with a lightweight CNN model, and dynamically decides on-the-fly whether to compute more powerful features for incoming video frames at a finer scale to obtain more details. This is achieved by a coarse LSTM and a fine LSTM operating cooperatively, as well as a conditional gating module to learn when to allocate more computation.

Available Sources:

- Official NeurIPS proceedings:
<https://proceedings.neurips.cc/paper/2019/hash/bd853b475d59821e100d3d24303d7747-Abstract.html>
- arXiv paper: <https://arxiv.org/abs/1912.01601>
- PDF: <http://papers.neurips.cc/paper/8993-liteeval-a-coarse-to-fine-framework-for-resource-efficient-video-recognition.pdf>

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-06-13T20:21:32.265Z)

Web search results for query: "arXiv 2507.06411 hierarchical multi-stage transformer context-aware temporal action localization PCL-Former"

Links: [{"title": "[2507.06411] Hierarchical Multi-Stage Transformer Architecture for Context-Aware Temporal Action Localization", "url": "https://arxiv.org/abs/2507.06411"}, {"title": "Hierarchical Multi-Stage Transformer Architecture for Context-Aware Temporal Action Localization", "url": "https://arxiv.org/html/2507.06411v1"}, {"title": "Hierarchical Multi-Stage Transformer Architecture for Context-Aware Temporal Action Localization | Request PDF", "url": "https://www.researchgate.net/publication/393539229_Hierarchical_Multi-Stage_Transformer_Architecture_for_Context-Aware_Temporal_Action_Localization"}, {"title": "Spatio Temporal Action Localization", "url": "https://www.catalyzex.com/s/Spatio%20Temporal%20Action%20Localization"}, {"title": "HIERARCHICAL MULTI-STAGE TRANSFORMER AR-", "url": "https://arxiv.org/pdf/2507.06411"}, {"title": "Temporal Action Localization in Untrimmed Videos via Multi-stage CNNs", "url": "https://www.researchgate.net/publication/311609422_Temporal_Action_Localization_in_Untrimmed_Videos_via_Multi-stage_CNNs"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers | Computer Vision ? ECCV 2022", "url": "https://dl.acm.org/doi/10.1007/978-3-031-19772-7_29"}]

Based on the search results, I found detailed information about arXiv paper 2507.06411:

PCL-Former: Hierarchical Multi-Stage Transformer for Temporal Action Localization

PCL-Former is a hierarchical multi-stage transformer architecture where each subtask is handled by a dedicated transformer module with a specialized loss function.

Architecture Overview

The architecture consists of three components: the Proposal-Former identifies candidate segments in untrimmed videos that may contain actions, the Classification-Former classifies the action categories within those segments, and the Localization-Former precisely predicts the temporal boundaries (start and end) of the action instances.

Performance

The PCL-Former outperforms state-of-the-art temporal action localization (TAL) approaches by 2.8% on THUMOS14, 1.2% on ActivityNet-1.3, and 4.8% on HACS datasets.

Evaluation

The method was evaluated on three challenging benchmark datasets: THUMOS-14, ActivityNet-1.3, and HACS Segments.

The paper was submitted to arXiv in July 2025 and explores the spatio-temporal properties of transformers within a multi-stage architecture specifically designed for the task of identifying and localizing actions within untrimmed video sequences.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-13T20:21:44.179Z)

All six citations now verify as REAL papers:

- arXiv:2507.06411 exists with the exact title described. The "PCL-Former" name (Proposal/Classification/Localization Formers) is confirmed by the abstract. The survey honestly flagged that authorship wasn't independently verified ? and indeed the survey did not assert specific author names, so no fabrication.
- LiteEval (arXiv:1912.01601, NeurIPS 2019) verifies with correct lead author Zuxuan Wu.

All citations check out. Now the adversarial core: the verdict is "partial-overlap," which is the conservative middle. The survey's own rationale concedes the hierarchical-context idea is

NOT novel and identifies only "a thin slice that may be a small genuine gap" ? namely the exact "scope-taxonomy-as-feature-columns" design with a tiny logistic scorer over hand-crafted multi-scale features, persist-once/re-score-forever, report-first A/B-gated reconcilers.

My job is to test whether that thin differentiated slice is actually as differentiated as claimed, or whether closer prior art downgrades it toward "reinventing." Let me hunt for the strongest refutations: (1) hand-crafted multi-scale temporal features + simple/logistic classifier for sports rally/play segmentation, and (2) coarse session-state-as-soft-feature conditioning fine decisions.

13. user (2026-06-13T20:21:51.922Z)

Web search results for query: "badminton rally segmentation shuttlecock trajectory handcrafted features classifier temporal IoU evaluation"

Links: [{"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.mdpi.com/1424-8220/24/13/4372"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}, {"title": "FineBadminton: A Multi-Level Dataset for Fine-Grained Badminton Video Understanding", "url": "https://arxiv.org/html/2508.07554v1"}, {"title": "Future Prediction of Shuttlecock Trajectory in Badminton Using Player's Information", "url": "https://www.mdpi.com/2313-433X/9/5/99"}, {"title": "One-Shot Badminton Shuttle Detection for Mobile Robots", "url": "https://arxiv.org/html/2603.06691v2"}, {"title": "Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos", "url": "https://arxiv.org/pdf/2002.08097"}, {"title": "VideoBadminton: A Video Dataset for Badminton Action Recognition | Request PDF", "url": "https://www.researchgate.net/publication/388099044_VideoBadminton_A_Video_Dataset_for_Badminton_Action_Recognition"}]

I found several relevant research papers and studies related to your search query on badminton rally segmentation and analysis. Here are the key findings:

Temporal IoU Evaluation in Badminton Analysis

Temporal Intersection over Union (t-IoU) is used as a metric for evaluating shot extraction performance in badminton game analysis. TrackNet, a deep neural network designed for tracking small objects, is used to extract the flight trajectory of the shuttlecock.

Handcrafted Features and Classifiers

A hybrid framework combines handcrafted spatiotemporal features with a two-stage feature selection and weighted ensemble learning, extracting RGB-based descriptors (Histogram of Oriented Gradients (HOG), Histogram of Optical Flow (HOF), Motion Boundary Histogram (MBH)) and skeleton-based features (Range of Motion Index (ROMI) for spatial and Dynamic Time Warping (DTW) for temporal).

Rally Segmentation and Shuttlecock Tracking

Automating rally segmentation and shuttlecock position detection in match videos is considered necessary for practical scenarios. TrackNet, a deep neural network designed for tracking small objects, is used to extract the flight trajectory of the shuttlecock, and the YOLOv7 model is used to identify whether the player is swinging.

One proposed method achieves an accuracy of 89.7%, a recall rate of 91.3%, and an F1 rate of 90.5% in 69 matches, with 1582 rallies of Badminton World Federation (BWF) match videos.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. user (2026-06-13T20:21:53.902Z)

Web search results for query: "tennis rally detection play segmentation handcrafted features logistic regression multi-scale temporal window untrimmed video"

Links: [{"title": "RallyClip | Free Tennis Match Segmentation", "url": "https://ioblesrazzaq.github.io/RallyClip/"}, {"title": "Predicting tennis match outcomes mid-game using machine learning on psychological and physical data | Journal of Big Data | Full Text", "url": "https://journalofbigdata.springeropen.com/articles/10.1186/s40537-025-01216-4"}, {"title": "Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos", "url": "https://arxiv.org/pdf/2002.08097"}, {"title": "Predicting tennis match outcomes mid-game using machine learning on psychological and physical data | Journal of Big Data | Springer Nature Link", "url": "https://link.springer.com/article/10.1186/s40537-025-01216-4"}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "Automatic rally detection and scoring", "url": "https://image-pubs.uspto.gov/dirsearch-public/print/downloadPdf/10751601"}, {"title": "TennisVid2Text: Fine-grained Descriptions for Domain Specific Videos", "url": "https://arxiv.org/pdf/1511.08522"}]

Based on the web search results, I found several relevant papers and projects related to tennis rally detection and video segmentation:

Key Findings:

One approach to rally detection in untrimmed tennis videos uses a sliding window method applied to the entire length of the input video, which provides contiguous confidence regions in time depicting the duration of a single rally, with rising and falling edges detected as serve points and end-of-rally points.

An alternative method employs a lightweight, purely audio-based detection model that trains a neural network to identify the distinct acoustic signatures of racquet-ball impacts from the broadcast audio track. Once hitting moments are identified, these temporal anchors are clustered in the temporal domain to form candidate rally intervals, with specific filtering rules applied to retain valid segments corresponding to complete, uninterrupted rallies.

The search results also include references to tennis video analysis systems that use various feature extraction and classification approaches. Logistic Regression, as a statistical direct probability model, offers advantages in handling binary classification problems by estimating the probability of an event occurrence by fitting data to a logistic curve.

For more specific information about handcrafted features and multi-scale temporal window approaches, you may want to examine the arxiv paper on "Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos" (arxiv.org/pdf/2002.08097), which appears to directly address rally detection methodologies.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

15. assistant (2026-06-13T20:22:01.929Z)

These are highly relevant. Two findings stand out as potential prior-art that the survey missed entirely:

1. **"Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos" (arXiv:2002.08097)** ? appears in BOTH searches, directly addresses rally detection in unstructured (non-broadcast) sports video. This could be very close prior art for the "static-tripod / no broadcast cues" gap the survey claims the literature doesn't address.
2. **The MDPI Sensors 2024 badminton paper (PMC11244353)** ? uses TrackNet shuttle trajectory + YOLOv7 swing detection + temporal IoU for rally segmentation on monocular camera. This is strikingly close to "our_pattern": frame-scope shuttle detector + person/swing cue, evaluated by temporal IoU.

Let me dig into both to assess how close they really are.

16. user (2026-06-13T20:22:15.572Z)

I appreciate your detailed request, but the provided content is a PDF file encoded in binary format (with FlateDecode compression). The actual paper text is not readable in the format you've shared?it contains only PDF structural metadata and compressed image/font data, not extractable text content.

To properly summarize the paper's method for event/rally detection, I would need:

- The paper in plain text or readable format, or
- A direct link to access it through an academic repository

If you can share the paper text or a clearer version, I'll be happy to address your five specific questions about the detection approach, temporal segmentation strategy, sports domain, and evaluation metrics using appropriately attributed quotes (?125 characters each).

[Binary content (application/pdf, 4MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\7d6bbb83-42c0-459b-9e23-e14e96d4cb2d\tool-results\webfetch-1781382128851-aul44a.pdf]

17. user (2026-06-13T20:22:15.573Z)

Summary: Badminton Shot Detection Method

(1) Camera Type

Yes, the method uses **monocular (single) camera footage**. The authors explicitly state: "The proposed method only analyzes videos captured from a monocular camera, making it applicable to existing badminton videos without any additional hardware." They test on both official BWF YouTube videos and competition footage?not broadcast-exclusive systems.

(2) Rally Segmentation via Multiple Cues

The method combines **two detection streams**:

- **TrackNet** (shuttlecock trajectory tracking)
- **YOLOv7** (player swing/hit action detection)

The Shot Refinement Algorithm (SRA) merges these: "Hit Detection by Trajectories (HD-Tt)" and "Hit Detection by Action (HD-A)" are combined to "provide fine-grained results." This dual-cue approach addresses detection misses in each modality independently.

(3) Dead Time / Between-Rally Detection

The paper does **not explicitly detect warm-up or non-match segments**. It segments footage into Hit Segments based on player action detection, but doesn't distinguish match vs. practice play or describe explicit gap-filling logic for idle periods between rallies.

(4) Classifier Type

The method uses **XGBoost**, a simple learned classifier: "limited by the number of features and samples available for training, this study employs the XGBoost model." Features are 7-dimensional (player ankle positions, flight duration, shuttlecock displacement).

(5) Evaluation Metric

Temporal Intersection over Union (t-IoU): $t\text{-IoU} = \frac{HMi(1?2) \cap GT}{HMi(1?2) \cup GT}$, where i is the i th hit moment, and GT is the groundtruth shots." A detected shot qualifies as true if t-IoU exceeds a threshold (tested at 0.5, 0.85, 0.95).

18. assistant (2026-06-13T20:22:22.380Z)

Very important findings. The MDPI Sensors 2024 badminton paper (PMC11244353) is strikingly close to "our_pattern": monocular non-broadcast footage, TrackNet shuttle trajectory + YOLOv7 swing/person cue fused, low-dimensional handcrafted aggregate features (7-d), a simple learned classifier (XGBoost), evaluated by temporal IoU. This is a much closer analogue to the actual fusion architecture than ANY of the six cited deep end-to-end action-segmentation papers ? and it's entirely absent from the survey. It directly undercuts the "static-tripod transfer gap" pitfall claim that "the literature won't have done that for you."

Let me fetch the arXiv:2002.08097 paper via the HTML/abstract route since the PDF didn't parse, and run one more search on the specific "coarse session-state as soft feature conditioning" claim to see if that's also reinvented.

19. user (2026-06-13T20:22:29.882Z)

Web search results for query: ""Unsupervised Temporal Feature Aggregation" event detection unstructured sports videos badminton tennis rally static camera"

Links: [{"title": "Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/8959041/"}, {"title": "Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos for ISM 2019 - IBM Research", "url": "https://research.ibm.com/publications/unsupervised-temporal-feature-aggregation-for-event-detection-in-unstructured-sports-videos"}, {"title": "[2002.08097] Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos", "url": "https://arxiv.org/abs/2002.08097"}, {"title": "Unsupervised Temporal Feature Aggregation for", "url": "https://arxiv.org/pdf/2002.08097"}, {"title": "A Comprehensive Review of Computer Vision in Sports: Open Issues, Future Trends and Research Directions", "url": "https://arxiv.org/pdf/2203.02281"}, {"title": "Event detection in soccer videos using unsupervised learning of Spatio-temporal features based on pooled spatial pyramid model | Multimedia Tools and Applications | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s11042-018-7083-1"}, {"title": "Action Spotting and Precise Event Detection in Sports: Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v1"}, {"title": "Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v3"}, {"title": "Learning sports highlights using event detection", "url": "https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/8923607"}]

I found the research paper you're looking for. The paper "Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos" addresses automatic retrieval of key events in sports games and studies event detection in unstructured environments with arbitrary camera angles.

The work identifies and solves two major problems: unsupervised identification of players in an unstructured setting and generalization of trained models to pose variations due to arbitrary shooting angles, proposing a temporal feature aggregation algorithm using person re-identification features to obtain high player retrieval precision.

The paper demonstrates this approach on table tennis videos, performing temporal feature aggregation and clustering-based player retrieval with data augmentation using a multi-modal image translation model, with training performed on augmented data to learn a rally detector.

The research was published in 2019 at the IEEE International Symposium on Multimedia (ISM 2019) by authors including Subhajit Chaudhury and colleagues from IBM Research and other institutions.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

20. user (2026-06-13T20:22:31.909Z)

Web search results for query: "warm-up versus match detection sports video gameplay cadence rally rhythm coarse state classification amateur footage"

Links: [{"title": "Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v3"}, {"title": "Automatic rally detection and scoring", "url": "https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10751601"}, {"title": "Action Spotting and Precise Event Detection in Sports: Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v1"}, {"title": "Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos", "url": "https://arxiv.org/pdf/2002.08097"}, {"title": "An overview of Human Action Recognition in sports based on Computer Vision - ScienceDirect", "url": "https://www.sciencedirect"}]

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.com/science/article/pii/S2405844022009215"}],{"title":"Building Scalable Video Understanding Benchmarks through Sports","url":"https://arxiv.org/pdf/2301.06866"}, {"title":"How AI-based computer vision algorithms are impacting the sports industry: a survey | Discover Artificial Intelligence | Springer Nature Link","url":"https://link.springer.com/article/10.1007/s44163-025-00586-1"}]]
```

I found relevant research on this topic. Here's what the search results show:

Key Findings

Sports Video Event Detection and Classification: A common limitation across most sports datasets is their reliance on professional broadcast footage, while amateur matches have different camera placement, video quality, and gameplay dynamics.

Rally Detection: Temporal segmentation evaluation is tailored to find how well detectors localize rally detection in time using sliding window methods that provide contiguous confidence regions depicting rally duration, with rising and falling edges detected as serve points and end-of-rally points.

Action Recognition in Sports: Human Action Recognition is used in sports such as volleyball, basketball, soccer, and tennis to detect players and recognize their actions during training, matches, warm-ups, or competitions. Research has specifically addressed outdoor footages of amateur tennis players for action classification tasks.

Match State Classification: Match state extraction involves detecting bounding boxes that encapsulate match state information in sports videos.

The search results indicate that distinguishing between warm-up and match gameplay, particularly with amateur footage, remains an area where researchers acknowledge the gap between professional broadcast standards and real-world conditions.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

21. assistant (2026-06-13T20:22:38.546Z)

This confirms the picture. arXiv:2002.08097 (Chaudhury et al., IBM Research, ISM 2019) is a real paper doing rally detection in unstructured (arbitrary-angle, non-broadcast) table-tennis video ? directly relevant to the "static tripod / no broadcast cues" gap. The second search confirms the gap framing is partly acknowledged in the literature (amateur-footage gap is a known limitation) but no single paper does the exact warm-up-as-soft-feature conditioning ? so that specific slice remains thin/genuine, consistent with the survey.

Let me do one final targeted check on the "warm-up/practice detection conditioning rally decisions" sub-claim, since that's the single most specific novel element the survey identified, to confirm no exact prior art exists.

22. user (2026-06-13T20:22:47.348Z)

Web search results for query: "automatic practice rally versus game point detection badminton tennis continuous rallying drill segmentation timeline"

Links: [{"title":"Automatic rally detection and scoring","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10751601"}, {"title":"TennisExpert: Towards Expert-Level Analytical Sports Video Understanding","url":"https://arxiv.org/pdf/2603.13397"}, {"title":"Towards Real-Time Analysis of Broadcast Badminton Videos","url":"https://arxiv.org/pdf/2308.12199"}, {"title":"A Real-Time Deep Learning Approach to Badminton Player Positioning and Movement Tracking | Request PDF","url":"https://www.researchgate.net/publication/395478974_A_Real-Time_Deep_Learning_Approach_to_Badminton_Player_Positioning_and_Movement_Tracking"}, {"title":"Automatic rally detection on broadcast tennis videos | Request PDF","url":"https://www.researchgate.net/publication/290310875_Automatic_rally_detection_on_broadcast_tennis_videos"}, {"title":"Service and End of Rally Detection in Badminton Videos | Springer Nature"}]

Link", "url": "https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15"}, {"title": "Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v3"}, {"title": "bfmd: a full-match badminton dense dataset for dense shot captioning", "url": "https://arxiv.org/html/2603.25533v1"}]]

I've performed the web search for your query about automatic practice rally versus game point detection in badminton and tennis. Here are the key findings:

Rally Detection Systems

Rally detection modules detect a rally in a play through of a sports game, where a rally is defined as a sequence of one or more strokes starting with a serve move. Rally detection uses the elapsed time between consecutive strokes and the stroke type to detect rallies.

Point/Rally Segmentation in Badminton

Broadcast badminton matches are divided into gameplays and replays, with "rally" frames defined as frames where actual gameplay is happening and "non-rally" frames as frames where replays or player rest periods occur. Recent methods have achieved 95.44% point segmentation accuracy on Olympic matches.

Service and Rally Endpoint Detection

The analysis requires annotating the start and end of the rally for each point, and techniques have been developed to detect the start of service and end of rally utilizing court details and poses to identify service actions. Experimental results show 88.6% accuracy for detecting rally points.

Temporal Segmentation Approaches

Temporal action localization is effective for analyzing long or continuous actions such as rallies in racket sports and is best applied to coarse temporal segmentation tasks, including highlight generation and detection of extended play phases.

These results cover automatic detection systems for distinguishing game/practice play periods and segmenting individual rallies in both sports.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

23. assistant (2026-06-13T20:23:00.906Z)

I have enough to finalize. No paper does the specific "warm-up/practice continuous-rally stretch detected as a session-scope span that conditions per-window rally decisions as a soft feature." The closest are broadcast rally/non-rally point segmentation (replay/rest based) and rally-endpoint detection ? different mechanisms. So that one specific slice remains a thin genuine gap, consistent with the survey.

Summary of my adversarial assessment:

****Citations:**** All six are REAL and accurately characterized. The two the survey self-flagged as not-fully-verified (PCL-Former arXiv:2507.06411, and it being careful not to assert GL-MSTCN/PCL-Former author lists) both check out ? arXiv:2507.06411 exists with the exact title and the PCL-Former three-stage description is confirmed. No fabrications.

****Novelty:**** The "partial-overlap" verdict holds, but the survey has a ****real coverage gap that weakens its own "what is genuinely differentiated" claim****: it missed the two closest analogues to the actual fusion architecture (not the deep end-to-end nets it cited). Specifically the MDPI Sensors 2024 badminton paper (PMC11244353) does monocular non-broadcast TrackNet-shuttle + YOLOv7-swing fusion ? low-dim handcrafted features ? simple learned classifier (XGBoost) ? temporal-IoU eval, and Chaudhury et al. ISM 2019 (arXiv:2002.08097) does rally detection in unstructured/arbitrary-angle (non-broadcast) sports video. These directly soften the survey's

"static-tripod transfer gap / the literature won't have done that for you" pitfall. The verdict stays partial-overlap (not reinventing) because neither does the persist-once/re-score-forever, calibrated (value,confidence) feature-column regime, the report-first A/B-gated reconcilers, or the warm-up-as-soft-session-feature conditioning ? that specific slice remains a thin genuine gap.

24. user (2026-06-13T20:23:47.486Z)

Output does not match required schema: root: must NOT have additional properties

25. assistant (2026-06-13T20:23:54.115Z)

I included an extra key. Let me resubmit with only the schema-defined properties.

26. user (2026-06-13T20:24:29.861Z)

Structured output provided successfully

27. assistant (2026-06-13T20:24:35.040Z)

Adversarial fact-check complete. All six citations verified REAL with no fabrications; corrected verdict remains partial-overlap, with a flagged coverage gap (two missed nearest-neighbor papers that soften the survey's static-tripod novelty framing).